

Intelligent Gas Cylinder Filling Safety & Pressure Monitoring System

Project Design Baseline — Revision 2

Independent automated safety layer for O₂ / N₂ cylinder filling

Industrial safety and pressure-monitoring system for O₂/N₂ cylinder filling. The system adds an independent automated safety layer alongside the operator and does not replace the operator's responsibility for cylinder approval.

ENGINEERING NOTE

120 bar and 125 bar are Company-Approved Setpoints for this project. They are not independently validated legal or engineering limits and must be checked against actual equipment specifications, applicable standards, and qualified technical/safety approval before installation.

REVISION LOG

Rev.	Section	Change
2	6, 11.7, BOM	Added mechanical relief valve (passive, independent of controller).
2	10, 14	Documented PG1 as retained independent visual reference.
2	SR-08, DI-03, BOM	Reset changed to physical key switch.
2	3, 13	Added basis for 120–125 bar margin (manual sequential fill rate).

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1. Real Process and Equipment

- The shop fills Oxygen (O₂) and Nitrogen (N₂).
- There are 8 large source cylinders, stated for this project as 60 L and 160 bar.
- Each source cylinder has its own hose/path and a separate manually operated valve V1–V8.
- Normally, two source-cylinder valves are never open simultaneously.
- The Ramp/Manifold is factory-made, standardized and government-approved; this project does not redesign the Ramp.
- The Ramp has one Main Valve leading to the filling hose, an existing analog pressure gauge PG1, and (added in this revision) a mechanical relief valve — see Section 6.
- The filling hose terminates in a filling clamp for the customer cylinder.
- Filling proceeds sequentially from the lower-pressure source cylinder toward the higher-pressure source cylinder; source cylinder #8 is stated to be above 120 bar and can complete the fill.
- The operator opens one source-cylinder valve, waits for pressure transfer/stabilization, closes it, then moves to the next.
- At approximately 120 bar, the operator completes the normal filling procedure and disconnects the customer cylinder.
- There is currently no automatic shutdown described for the manual process.

2. Cylinder Approval Boundary

- Before filling, the operator checks the test sheet and physical appearance.
- The operator remains solely responsible for deciding whether a cylinder is eligible for filling.
- The system does not inspect test dates, certification, corrosion, dents, physical condition or other approval criteria.
- The safety-system scope begins after the operator has approved the cylinder and filling has started.

3. Company-Approved Setpoints

Setpoint	Meaning	System Action
120 bar	Warning / Target Reached	Warning/target indication; not an emergency shutdown threshold.
125 bar	Emergency Shutdown	Initiate shutdown, close Smart Main Valve, disable filling and enter LOCKED/FAULT.

Basis for the 5-bar warning margin

The 120→125 bar span is adequate because the approach to final pressure is inherently slow: filling is performed manually and sequentially through up to 8 source cylinders, with the operator opening one valve at a time and waiting for pressure transfer/stabilization before moving to the next. There is no credible fast-ramp scenario in this manual sequencing pattern that would carry the system from 120 to 125 bar faster than the controller and Smart Main Valve can respond. This basis should be re-confirmed once actual valve closure time and worst-case flow rate are known.

4. Physical Process Architecture

Element	Status	Notes
C1–C8	8 Source Cylinders	60 L / 160 bar each, individually valved V1–V8
Ramp / Manifold	Existing, unmodified	Factory-approved; carries PG1 (analog gauge) and, in this revision, a mechanical relief valve
Smart Main Valve Assembly (SMVA)	New	Electric shutoff valve + pressure transducer + digital gauge, installed at the existing Main Valve location
Filling Hose → Filling Clamp	Existing, unmodified	Terminates at the customer cylinder

Flow path: C1–C8 (via V1–V8) → Common Ramp/Manifold (PG1 + relief valve, existing/added) → Smart Main Valve Assembly → Filling Hose → Filling Clamp → Customer Cylinder.

5. Smart Main Valve Assembly (SMVA)

The existing Main Valve location is replaced by a Smart Main Valve Assembly. The approved Ramp itself remains unchanged.

Component	Function
Electrically controlled high-pressure shutoff valve	Primary automated actuator; closes on emergency shutdown or valve fault
Pressure transducer	Electronic measurement for controller/safety logic
Digital pressure gauge	Operator-facing indication

- Automatic emergency shutdown capability.
- The company will provide suitable equipment for O₂/N₂ and high-pressure service.
- Exact brand/model selection remains for detailed component selection.
- **Oxygen service scope:** wetted-material compatibility, cleaning/degreasing standard, and ignition-risk considerations for O₂ service (adiabatic compression heating, particulate/contamination control) are addressed during detailed equipment selection, not in this control-logic document.

6. Defense-in-Depth Safety Architecture

UPDATED IN THIS
REVISION

A single electronic measurement-and-shutdown chain is not sufficient for a hazard rated Critical. Four layers are in place so that no single point of failure, mechanical, electronic, or human, can cause uncontrolled continuation of filling.

#	Element	Type	Trigger	Independence Rationale
1	Mechanical relief valve (NEW)	Passive / mechanical	Automatic	Independent of controller, power, and software. Vents on overpressure regardless of electronic system state.
2	Ramp gauge PG1 (existing, retained)	Mechanical / visual	Operator-observed	Non-electronic pressure reference with no shared failure mode with the transducer/controller/display chain. Lets the operator visually cross-check the digital display, especially during Warning state.
3	Pressure transducer + Safety Controller	Electronic	Automatic	Primary automated safety measurement and shutdown logic (120 bar warning, 125 bar emergency shutdown).
4	Physical E-Stop	Electromechanical	Manual, operator-initiated	Independent manual override through a dedicated safety circuit; forces shutdown and lockout.

Layers 1 and 2 added this revision. The relief valve was proposed as an offloading path independent of the electronic system; PG1 was confirmed retained per the project principle of adding new protection without modifying the existing approved Ramp.

Layer 3 is the only layer that depends on power, firmware, and a single sensor. Layers 1 and 2 share no failure mode with it: the relief valve is purely mechanical, and PG1 is a direct analog reading with its own sensing path. A silent transducer drift or controller fault removes the automated layer only; the mechanical and visual layers stay intact.

7. Hazard Analysis

Hazard	Initial Severity
Overpressure	Critical
Pressure gauge failure	Critical
Pressure sensor failure	Critical
Shutdown valve/device failure	Critical
Two paths accidentally open simultaneously	High
Wrong source-cylinder selection	High
Power failure	High
Sensor/cable failure	High
Abnormal pressure rise	High

Hazard	Initial Severity
Operator error	Critical

CORE SAFETY PRINCIPLE

Human operation remains present, but an automated safety layer monitors pressure and can force the filling path to a safe state. No single point failure should cause uncontrolled continuation of filling.

8. Safety Requirements Specification (SRS)

- **SR-01** — Continuously measure/monitor pressure.
- **SR-02** — Display current pressure to the operator.
- **SR-03** — Detect pressure increase/decrease.
- **SR-04** — At 120 bar activate Warning/Target Reached indication.
- **SR-05** — At ≥ 125 bar initiate Emergency Shutdown.
- **SR-06** — After Emergency Shutdown enter LOCKED/FAULT.
- **SR-07** — No automatic restart after pressure drops or controller restarts.
- **SR-08** — Reset requires intentional authorized action via a physical key switch; Reset does not automatically start filling.
- **SR-09** — Detect sensor disconnect, invalid/out-of-range signal, frozen/unresponsive signal and electrical fault.
- **SR-10** — Sensor failure must not permit uncontrolled filling; transition to safe state and disable filling.
- **SR-11** — Power loss must not cause unsafe continuation; exact hardware behavior is defined during detailed design.
- **SR-12** — Startup: Power ON $\rightarrow$ Self Check $\rightarrow$ READY $\rightarrow$ operator action $\rightarrow$ START. No auto filling.
- **SR-13** — Physical Emergency Stop causes safety action, disables filling and locks the system.
- **SR-14** — Log Start, Stop, Warning, Emergency Shutdown, Sensor Fault, Power Failure, Reset and Maximum Pressure.
- **SR-15** — (New) Mechanical relief valve on the Ramp provides passive overpressure protection independent of controller state, power availability, and software logic.

GOLDEN PRINCIPLE

No single point failure should cause uncontrolled continuation.

9. State Machine

State	Transition
POWER ON	$\rightarrow$ SELF CHECK
SELF CHECK	OK $\rightarrow$ READY • FAULT $\rightarrow$ LOCK
READY	START $\rightarrow$ FILLING
FILLING	$P < 120 \rightarrow$ FILLING (continues)
	$120 \leq P < 125 \rightarrow$ WARNING
	$P \geq 125 \rightarrow$ EMERGENCY $\rightarrow$ VALVE CLOSED $\rightarrow$ LOCK
	SENSOR FAULT $\rightarrow$ SAFE STATE $\rightarrow$ LOCK
	VALVE FAULT $\rightarrow$ SAFE RESPONSE $\rightarrow$ LOCK
	E-STOP $\rightarrow$ SHUTDOWN $\rightarrow$ LOCK
LOCK	Authorized RESET (key switch) $\rightarrow$ SELF CHECK $\rightarrow$ READY

No direct LOCK → FILLING transition. Reset only returns to READY; Start is required separately.

10. Safety and Monitoring Architecture

Safety layer

Pressure → Safety Measurement → Safety Logic → Shutdown.

Monitoring layer

Controller → Display / Alarm / Logging / UI.

Signal path

- Pressure Sensor → Safety Controller → Warning/Alarm, Digital Display, and Safety Shutdown → Filling Disabled → Lockout.
- Ramp gauge PG1 remains in place and unmodified, providing an independent, non-electronic pressure reference the operator can visually cross-check against the digital display. It is not removed or superseded by the SMVA.
- Safety layer handles dangerous-condition detection, emergency shutdown, fault handling and lockout.
- Monitoring layer handles pressure display, status, warning, logging and reporting.
- Monitoring software must not override safety shutdown.
- PC, internet or Python must not be the sole mechanism responsible for emergency shutdown.
- Physical Emergency Stop should operate through an appropriate safety circuit.

11. Hardware Architecture

11.1 Pressure Transducer

- Primary electronic pressure measurement.
- 4–20 mA is the current preferred concept, subject to actual equipment.
- Selection parameters: pressure range, accuracy, pressure rating, O₂/N₂ compatibility, wetted materials, oxygen-service preparation and diagnostics.
- Generic conversion: $P = P_{min} + ((I - I_{min}) / (I_{max} - I_{min})) \times (P_{max} - P_{min})$.
- Example only: with a 0–200 bar sensor, 4 mA = 0 bar, 20 mA = 200 bar, 12 mA = 100 bar. Actual range is TBD.

11.2 Digital Pressure Gauge

- Operator-facing display of pressure and status.
- Not the sole safety decision source; cross-checked against the retained analog gauge PG1 (Section 10).

```
GAS FILLING SYSTEM
PRESSURE: 116.8 bar
FILLING
VALVE: OPEN

Warning: 120.1 bar — TARGET REACHED
Emergency: 125.1 bar — VALVE CLOSED — SYSTEM LOCKED
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11.3 Safety Controller

- Industrial/safety-capable controller rather than a basic Arduino/Raspberry Pi as the sole safety controller.
- Analog/digital I/O, diagnostics, watchdog, safety-oriented control capability, valve control and E-stop integration.

11.4 Electric Shutoff Valve

- Replaces the existing Main Valve function at the same location.
- Must be appropriate and approved for high-pressure O₂/N₂ service.
- Valve position feedback is strongly preferred.

11.5 Valve Position Feedback

CLOSE command → Electric Valve → Position Feedback → Controller.

Command = CLOSE but feedback = OPEN → VALVE FAULT → Safety Response → LOCK.

11.6 Emergency Stop

E-STOP → Safety Circuit → Shutdown + LOCK → Valve Closed.

11.7 Mechanical Relief Valve

- Passive mechanical relief/safety valve mounted on the Ramp, independent of the electronic control path.
- Vents automatically on overpressure with no dependency on controller power, firmware, or logic state.
- Proposed during design review and accepted by the company; sizing and set pressure are pending equipment selection and qualified safety/engineering approval.

- Addition only, not a modification to the existing approved Ramp, consistent with the project's non-negotiable principle of not altering the approved Ramp design.

11.8 Power and Watchdog

- Control power: controller, sensor, display and indicators.
- Actuator power: electric shutoff valve.
- Power-failure behavior must prevent unsafe continuation and automatic restart.
- Watchdog should detect controller/firmware failure and initiate the defined safe response.

12. Preliminary I/O List

Analog Inputs

Tag	Signal	Function
AI-01	Pressure Transducer	Main pressure measurement

Digital Inputs

Tag	Signal	Function
DI-01	Emergency Stop Status	Emergency condition
DI-02	Start	Start filling
DI-03	Reset (key switch)	Reset lock/fault — authorized action
DI-04	Valve Open Feedback	Valve position
DI-05	Valve Closed Feedback	Valve position
DI-06	External Fault	Equipment diagnostic

Digital Outputs

Tag	Signal	Function
DO-01	Shutoff Valve Command	Valve control
DO-02	Warning Lamp	120-bar warning
DO-03	Buzzer	Audible warning
DO-04	Emergency Lamp	Emergency status
DO-05	Ready Indicator	Ready status
DO-06	Filling Indicator	Filling status

This I/O list is preliminary and will be updated after actual company-supplied equipment is selected. PG1 and the mechanical relief valve are passive/mechanical and are not wired into the controller I/O.

13. Pressure Logic

Condition	State
$P < 120 \text{ bar}$	NORMAL / FILLING
$120 \text{ bar} \leq P < 125 \text{ bar}$	WARNING / TARGET REACHED
$P \geq 125 \text{ bar}$	EMERGENCY SHUTDOWN → Valve Close → Filling Disabled → LOCK

At the current project definition, 120 bar is Warning/Target; 125 bar is Emergency Shutdown. The 5-bar margin is considered adequate given the slow, manually sequenced approach to final pressure (Section 3); this should be re-confirmed against actual valve closure time during detailed design.

14. Sensor Fault Logic

- Disconnected sensor.
- Invalid electrical signal.
- Out-of-range measurement.
- Frozen/stuck/unresponsive signal.
- Electrical fault.
- Where feasible, implausible pressure behavior.

Sensor Fault → Safe Response → Filling Disabled → LOCK.

The retained analog gauge PG1 remains available to the operator as a visual reference even during a transducer/controller sensor fault.

15. Valve Fault Logic

If feedback is available: Command ≠ Actual Position → VALVE FAULT → Safety Response → LOCK.

16. Power Failure

During filling: FILLING → POWER FAILURE → SAFE STATE → LOCK / FILLING DISABLED.

After restoration: POWER RESTORED → SELF CHECK → READY.

No automatic restart. The mechanical relief valve remains active during a power failure, since it does not depend on electrical power.

17. Event Logging

- System Start
- System Stop
- Filling Start
- Filling Stop
- 120-bar Warning
- Emergency Shutdown
- Sensor Fault
- Valve Fault
- Emergency Stop
- Power Failure
- Reset (with operator/key identifier if available)
- Maximum Pressure

Potential future fields: date/time, filling cycle ID, duration, fault cause, system state and operator ID if required.

18. Preliminary Bill of Materials (BOM)

No.	Component	Qty.
1	High-Pressure Electric Shutoff Valve	1
2	Pressure Transducer	1
3	Digital Pressure Gauge / Display	1
4	Safety Controller	1
5	Emergency Stop	1 or more
6	Warning Lamp	1
7	Buzzer	1
8	Reset Key Switch	1
9	Start Button	1
10	Status Indicators	1 set
11	Power Supply	1
12	Electrical Protection Components	1 set
13	Valve Position Feedback	1
14	Mechanical Relief / Safety Valve	1
15	Control Panel / Enclosure	1
16	Terminal / Wiring Components	1 set

Item 8 updated from generic 'Reset Button' to key switch; Item 14 (relief valve) added in this revision.

19. Project Phases and Status

Phase	Topic	Status
1	Process Understanding	Completed
2	Hazard Analysis	Initial version completed
3	Safety Requirements Specification	Initial version completed
4	System Architecture	Completed
5	Hardware Architecture	Updated — relief valve added
6	Circuit & Wiring Architecture	Conceptual design completed
7	Detailed Component Selection	Next
8	Detailed Safety Circuit	Next
9	Prototype Design	Later
10	Firmware	Later
11	Monitoring / Data Logging	Later
12	UI	Later
13	Testing & Validation	Later
14	Real Equipment Integration	Final stage
15	Documentation	Ongoing

20. Next Planned Step — Phase 6.2: Detailed Safety Circuit Design

- Detailed safety circuit.
- E-Stop path.
- Shutoff valve control path.
- Valve feedback path.
- Pressure transducer input.
- Mechanical relief valve placement and set-pressure coordination with the 125-bar electronic setpoint.
- Power architecture.
- Controller I/O mapping.
- Fault states.
- Reset path (key switch wiring).
- Alarm outputs.
- Terminal-to-terminal wiring documentation.

21. Prototype Validation Plan

Test Condition	Expected Result
119 bar	Normal
120 bar	Warning
124.9 bar	Warning
125 bar	Shutdown
Sensor disconnect	Fault
Controller restart	No auto-start
E-Stop	Shutdown
Reset (key switch)	Ready
Start	Filling

Only after logic validation should integration with actual high-pressure equipment be considered, subject to company and qualified technical/safety validation.

22. Non-Negotiable Project Principles

- Do not redesign the factory-approved Ramp.
- Replace the existing Main Valve function with the Smart Main Valve Assembly at the Main Valve location.
- Use company-provided, appropriately rated high-pressure/O₂/N₂ equipment.
- Operator remains solely responsible for cylinder eligibility/approval.
- Safety-system scope starts once filling begins.
- Monitoring software cannot override safety shutdown.

- No automatic restart after shutdown, fault or power restoration.
- Reset $\neq$ Start; reset is performed via a physical key switch.
- 120 bar = Company-Approved Warning / Target.
- 125 bar = Company-Approved Emergency Shutdown.
- Safety-critical logic should be independent from PC/internet/UI as practical.
- Valve position feedback is strongly preferred.
- Pressure transducer is the safety measurement source; digital gauge is primarily operator indication.
- **(New)** Existing Ramp gauge PG1 remains in place and unmodified, serving as an independent, non-electronic pressure reference for the operator.
- **(New)** A mechanical relief valve is added to the Ramp as a passive overpressure protection layer, independent of the electronic control path; an addition, not a modification, to the existing approved Ramp.
- Final installation requires validation against actual equipment, applicable standards and qualified safety/engineering approval.

End of Design Baseline, Revision 2. Subject to detailed component selection and qualified safety/engineering approval.